

# AN2DL - Second Challenge Report

## Augmented Whales

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## 1 Introduction

The objective of this challenge was to **classify diseased human tissue** samples. We had to predict the correct molecular subtype of each sample based on tissue data.

### 1.1 Approach Overview

We approached this image classification problem by doing these steps:

**Dataset Analysis and Data Preprocessing:** understand the data, look for outliers and prepare the data for modeling;

**Model Selection and Evaluation:** choose the best deep learning architecture for the task and assess the model's effectiveness using appropriate metrics.

## 2 Problem Analysis

### 2.1 Dataset characteristics

The dataset consists of tissue samples with labels indicating their molecular subtypes. Each sample is represented by an RGB image along with a binary mask highlighting the region of interest.

We immediately identified many outliers in the dataset and we removed them.



Figure 1: Example of outliers found in the dataset.

We also noticed a class imbalance, in particular **Triple Negative** class was underrepresented compared to the others.

### 2.2 Main Challenges

The main challenges we faced were:

- **Data Quality:** Images were very different in terms of resolution and size, requiring careful preprocessing to ensure consistency.
- **Class Imbalance:** The uneven distribution of classes necessitated the use of techniques to mitigate bias towards the majority classes.
- **Model Complexity:** Selecting an appropriate model architecture that balances complexity and generalization was crucial to avoid overfitting.
- **Reproducibility:** Making sure that all of our augmentations and tiles extraction were reproducible

### 3 Methods

#### 3.1 Data preprocessing

To avoid losing important features, we decided not to resize the images, but to crop them around the region of interest.

We implemented a **region-based** approach that identifies tumor regions using connected component analysis on the binary masks. **Connected component analysis** identifies contiguous regions of pixels with the same value (in this case, the tumor area). Tiles of  $224 \times 224$  pixels are extracted centered on these regions. This method ensures that extracted tiles focus on relevant areas rather than background tissue.

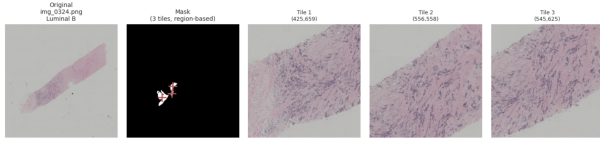


Figure 2: Example of tile extraction.

#### 3.2 Data Augmentation

To enhance the model’s robustness and generalization capabilities, we applied various data augmentation techniques during training. Specifically, we implemented a custom pipeline which includes:

- **Geometric Transformations:** To leverage cellular rotational symmetry, we applied random flips, discrete 90-degree rotations.
- **Color and Intensity Variations:** We applied moderate color jittering, randomizing brightness, contrast, saturation, and hue.

To keep consistency between the images and the relative masks, we implemented a **synchronized augmentation pipeline**. Geometric transformations are applied identically to both the RGB image and the mask to ensure spatial alignment is preserved, while color and intensity Variations are applied only to RGB images. Finally, the RGB channels are normalized using ImageNet statistics, while the mask remains unchanged.

#### 3.3 Data Normalization

Following augmentation, pixel values were normalized to match the statistics of the pre-trained back-

bone. We used the standard **ImageNet normalization parameters**.

#### 3.4 Model characteristics

We experimented with several deep learning architectures, starting from simple Convolutional Neural Networks (CNNs) to more complex models using parallel processing.

Our final model is based on a **dual path architecture** that combines features from two different CNN backbones: **DenseNet121** and **EfficientNetV2**. The outputs of both networks are concatenated and passed through fully connected layers to produce the final classification.

The first path uses DenseNet121 to extract **shape features** from the mask, while the second path employs EfficientNetV2 to capture **texture and color** information from the RGB images. This combination allows the model to get complementary features for improved classification performance.

#### 3.5 Training and Validation

Before extracting tiles, we split the dataset into training and validation sets in order to keep samples from the same patient in the same set.

The model was trained using the **Lion** optimizer with a learning rate of  $2e-5$  and a batch size of 32. We employed early stopping based on validation loss to prevent overfitting.

We used a **weighted focal loss** function to address class imbalance, assigning higher weights to underrepresented classes.

For validation, we evaluated model performance using **logits aggregation**: for each patient, we averaged the logits across all tiles before computing the prediction. This provides more robust patient-level classification than simple majority voting.

#### 3.6 Submission generation

Once we were satisfied with the model’s performance, we performed inference over the test set using the same tile extraction and logits aggregation strategy as in validation. Then we generated a **.csv** file with the predictions.

Table 1: Main experiments.

| Model                | Architecture | Image Usage      | Mask usage       | F1 val        | F1 sub        |
|----------------------|--------------|------------------|------------------|---------------|---------------|
| Baseline             | Sequential   | Resize           | 4 channels input | 0.3098        | 0.3259        |
| Intermediate Version | Sequential   | Tiles extraction | No usage         | 0.4114        | 0.4120        |
| Final                | Parallel     | Tiles extraction | Used in 1 path   | <b>0.4442</b> | <b>0.4359</b> |

## 4 Experiments

To identify the optimal model configuration, we conducted several experiments, with the key parameters and results summarized in Table 1.

### 4.1 Baseline

Our baseline model was a simple CNN architecture based on EfficientNetV2. We trained it without tiles extraction: we simply resized all images to a fixed size of 224x224 pixels and added a fourth channel to include the binary mask. We added a few augmentation techniques like random rotations and flips.

We trained the whole model without freezing any layers and obtained a validation F1-score of **0.30**. We also tried to finetune the pretrained model by freezing the backbone and training only the last layers, but the performance was even worse.

### 4.2 Tiles Extraction

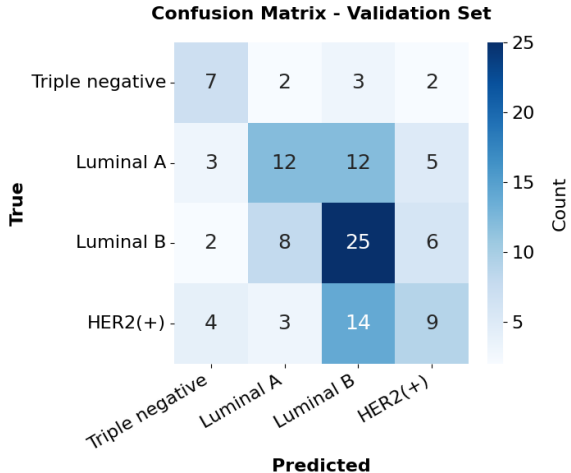
At this point we implemented the **tiles extraction** method described in Section 3.1. We also introduced a more advanced optimizer (Lion) and a weighted focal loss function to address class imbalance. We also changed model and used DenseNet121 as backbone. A more aggressive augmentation was also introduced at this stage to improve model generalization. Here we used only the RGB channels as input, since the mask was used to get the region of interest for tiles extraction.

This setup got a validation F1-score of **0.4114**, a significant improvement over the baseline.

### 4.3 Final Model

Finally, we implemented the dual path architecture described in Section 3.4, combining DenseNet121 and EfficientNetV2 backbones, keeping all the previous improvements.

This final model achieved a validation F1-score of **0.4442**, showing the effectiveness of combining complementary features from both backbones.



## 5 Results

The final dual-path model achieved a submission F1-score of **0.4359** over the test set. Region-based tile extraction and weighted focal loss contributed significantly to this achievement, along with patient-level logits aggregation, which led to more robust predictions compared to majority voting.

## 6 Discussion

The results confirm the benefit of combining shape features and texture/color information from masks and RGB images through a dual-path architecture. However, the overall F1-score remains limited, likely due to class imbalance and high variability across samples. More advanced aggregation or attention-based methods could further enhance performance.

## 7 Conclusions

We proposed a deep learning pipeline for molecular subtype classification that addresses data heterogeneity and class imbalance through tile-based preprocessing and a dual-path CNN model. The observed improvements over simpler approaches indicate that this strategy is effective and suitable for challenging histopathological classification tasks.